

Credit - Drishti Ias

India-Uzbekistan Comprehensive Strategic Partnership



For Prelims: Uzbekistan , Dustlik military exercise , World Trade Organization , , International Solar Alliance , Chabahar Port , International North-South Transport Corridor

For Mains: India-Central Asia relations, India's strategic partnerships and regional diplomacy

Source: PIB

Why in News?

The Prime Minister undertook a State Visit to **Uzbekistan** to commemorate the **15th anniversary of the India-Uzbekistan Strategic Partnership (2011)**. During the visit, the two nations elevated their bilateral relations to a **Comprehensive Strategic Partnership** .

- The visit comes at a time when stakes are high for India as it actively seeks to reset and deepen its ties with **Central Asian countries** amid shifting geopolitical dynamics.



Summary

- **India and Uzbekistan have upgraded their ties to a Comprehensive Strategic Partnership** , expanding cooperation in **defence, trade, critical minerals, uranium, clean energy, digital technology, healthcare and culture** .
- **Uzbekistan is crucial to India's Connect Central Asia policy** due to its strategic location and role in regional connectivity. India must address **connectivity, China's influence, trade barriers and security challenges** while leveraging **Chabahar, DPI, critical minerals and development cooperation** .

What are the Key Outcomes of the PM Visit to Uzbekistan?

- **Highest Civilian Honour:** The Prime Minister Narendra Modi was conferred with the "Oliy Darajali Do'stlik" Order, Uzbekistan's highest state honour.
- **Comprehensive Strategic Partnership:** Both nations formally upgraded their bilateral relationship to a **Comprehensive Strategic Partnership** , broadening long-term engagement across strategic, economic, and technological domains.
- **Defence, Security & Counter-Terrorism:** Both countries adopted a zero-tolerance approach to terrorism and called for action against terror financing, safe havens and perpetrators.
 - Both sides agreed to expand **defence-industrial cooperation** through the **Joint Working Group on Military Cooperation** .
 - They welcomed the successful **Dustlik military exercise** , held in Uzbekistan in April 2026.
- **Trade, Investment & Economic Integration:**
 - **Trade:** Set a bilateral trade target of **USD 5 billion by 2030, up from the current volume of over USD 1 billion**.
 - Both Nations agreed to establish a Coordination Council led by Foreign Ministers to oversee bilateral cooperation and upgrade the existing Joint Commission from the secretarial to the ministerial level.
 - **Non-Tariff Barriers:** Created a **Joint Working Group** to address trade and investment barriers in sectors such as **engineering, automobiles, electrical goods and pharmaceuticals** .
 - **Preferential Trade Agreement (PTA):** Both sides agreed to take **time-bound steps towards an India-Uzbekistan PTA** .
 - **WTO:** India reaffirmed its support for **Uzbekistan's** accession to **World Trade Organization (WTO)**.
 - **Investment:** Cooperation between Invest India and Uzbekistan's Investment Promotion Agency will promote two-way investment flows.
- **Critical Minerals, Energy & Nuclear Cooperation:**
 - **Critical & Strategic Minerals Supply Chain:** Agreed to implement joint ventures in geological exploration, mining, processing, value addition, and recycling of **critical minerals** .
 - The two nations agreed to establish a framework for the long-term supply of **uranium** from Uzbekistan to India to meet the fuel requirements of **India's civil nuclear energy programme**.

- **Civil Nuclear & Clean Energy:** Explored avenues for extended cooperation in the **peaceful uses of nuclear energy** , modernization of energy grids, and renewable energy infrastructure.
- **Digital Transformation:** Decided to scale up partnerships between startups, tech incubators, and IT parks, building on the success of the **Tashkent IT Park (established in 2019 with [Software Technology Parks of India\(STPI\)](#) assistance)**.
 - Agreed to collaborate on developing secure and trustworthy Artificial Intelligence (AI) systems for economic development and societal problem-solving.
- **Digital Public Infrastructure:** An agreement between **[NPCI International Payments Ltd](#)** and Uzbekistan's National Interbank Processing Centre will allow Indian **UPI apps to scan Uzbekistan's UZQR** national codes, boosting digital connectivity.
- **Aral Sea Ecological Restoration:** India announced a **USD 1 million grant** under its Grant-in-Aid framework to support afforestation and ecological rehabilitation in the fragile Aral Sea region, linking Uzbekistan's "*Yashil Makon*" with **India's "*Ek Ped Maa Ke Naam*"** .
- **Pharmaceuticals & Healthcare:** Agreed to utilize the **Tashkent Pharma Park** cluster for joint manufacturing, technology transfer, and establishing Indian medical laboratories and research centres in Uzbekistan.
- **Heritage Preservation & Buddhist Sites:** Signed a Letter of Intent to conserve and restore the ancient Buddhist heritage sites of **Fayaz Tepa** and **Karatepa** , along with a **Cultural Exchange Program (2026-2030)** .
- **Education & Traditional Knowledge:** India announced the **Lal Bahadur Shastri Scholarship** for Hindi language learning in Uzbekistan and signed an MoU for cooperation in **[traditional medicines \(AYUSH\)](#)**.
- **Multilateral & Regional Alignment:** Uzbekistan reaffirmed its support for India's candidature for permanent membership in a reformed **[UN Security Council](#)**.
 - Uzbekistan expressed its intent to join the **[International Solar Alliance \(ISA\)](#)** and the **[International Big Cat Alliance \(IBCA\)](#)** .
 - Uzbekistan backed India's **[2026 BRICS Chairship](#)** , while India pledged support for Uzbekistan's upcoming **[Non-Aligned Movement \(NAM\) Chairship \(2027-2030\)](#)** .
 - Both reaffirmed cooperation within the **[SCO](#)** and the **Central Asia-India** framework.

Click here to Read: [Recalibrating India's Central Asia Strategy](#)

Why is Uzbekistan Strategically Important to India?

- **Geopolitical Pivot of Central Asia:** Uzbekistan is at the **"heart" of Central Asia** , sharing borders with all other Central Asian Republics (Kazakhstan, Kyrgyzstan, Tajikistan, and Turkmenistan) as well as Afghanistan. It acts as the linchpin for India's **["Connect Central Asia Policy"](#)**.
- **Energy and Mineral Security:** Beyond uranium for nuclear energy, Uzbekistan possesses abundant reserves of gold, natural gas, and critical minerals essential for India's transition to clean energy technologies.
- **Countering Regional Threats:** The Taliban takeover in neighboring Afghanistan has heightened the shared risk of radicalization, drug trafficking, and cross-border terrorism.
 - Collaborative intelligence sharing and joint exercises (like **Exercise Dustlik**) act as a buffer against these threats.

- **Balancing China's Influence:** Through infrastructure investments and digital diplomacy (like UPI integration), India is offering a transparent, capacity-building alternative to China's debt-heavy Belt and Road Initiative (BRI) in the region.
- **Unlocking Connectivity:** Since India lacks direct land access to Central Asia, partnering with Uzbekistan is crucial for operationalizing trade transit routes, notably the **Chabahar Port (Iran)** and the **International North-South Transport Corridor (INSTC)**.

India-Uzbekistan Relations

- **Historical & Cultural Ties:** Relations have roots in the **Silk Route**, with the spread of **Buddhism** to Central Asia and historical links through **Babur**, founder of the Mughal Empire, who originated from the Fergana Valley in modern-day Uzbekistan.
 - India established diplomatic relations with Uzbekistan in **March 1992**. Bilateral ties were upgraded to a **Strategic Partnership in 2011**.
- **Defence Cooperation:** Both countries conduct **Exercise DUSTLIK** to enhance military interoperability, particularly in **counter-terrorism operations**.
- **State-to-Region Cooperation:** **Gujarat** has partnered with Uzbekistan's **Samarkand region**, reflecting growing sub-national diplomacy.
- **Trade & Investment:** Trade volume reached **USD 980 million in 2024**, with Indian exports led by pharmaceuticals, machinery, auto components, and mobile phones, while imports consist mainly of fruit/vegetable products, fertilizers, and lubricants.
- **Development Partnership:** India supports Uzbekistan through **High Impact Community Development Projects (HICDPs)**, including healthcare, IT and education initiatives.

What are the Key Challenges in India-Uzbekistan Relations?

- **Connectivity Deficit:** India has **no direct overland access** to Uzbekistan due to Pakistan's transit constraints, while Uzbekistan itself is **doubly landlocked**. Routes through **Chabahar Port and INSTC** increase transit time and freight costs.
- **Uranium Transit Risks:** Although long-term **Uzbek uranium supply** is a major opportunity, transporting nuclear material through **Central Asia and Iran** involves complex logistics, safety requirements and transit risks.
- **Banking & Payment Barriers:** Limited **direct banking and correspondent banking links** and the thinly traded **Uzbek Som** increase currency-conversion and settlement costs, particularly affecting Indian SMEs.
- **China's Economic Dominance:** China is Uzbekistan's **largest trading partner and major infrastructure financier** through the **BRI**. The **China-Kyrgyzstan-Uzbekistan (CKU) railway** could further improve China's logistical advantage over India.
- **Regional Security Risks:** Instability in **Afghanistan**, including radicalisation, drug trafficking and threats from extremist groups such as **ISIS-K**, creates security concerns for trade and connectivity projects.
- **Non-Tariff & Regulatory Barriers:** **Bureaucratic procedures, language barriers and complex**

non-tariff barriers restrict market access. A **Joint Working Group** has been established to address these obstacles and facilitate private-sector engagement.

What Measures are Needed to Strengthen India-Uzbekistan Relations?

- **Subsidizing Freight & Air Corridors:** Until land and sea routes become highly efficient, both governments should consider establishing subsidized "Air Freight Corridors" (similar to the one with Afghanistan) specifically for high-value, low-volume goods like pharmaceuticals, IT components, and agricultural produce.
- **Fast-Tracking the Preferential Trade Agreement (PTA):** With the joint feasibility study already concluded, an early conclusion of a PTA is vital to lower tariffs on core Indian exports (medicines, textiles, engineering goods) and Uzbek exports (fertilizers, fruits, lubricants).
- **Tackling Non-Tariff Barriers (NTBs):** Indian exporters face stringent and often opaque regulatory hurdles in Uzbekistan. A mutual recognition agreement (MRA) for product standards—especially in pharmaceuticals and agricultural phytosanitary checks—is urgently required.
- **Green Energy Investments:** As Uzbekistan seeks to modernize its energy grid, Indian solar companies should leverage the **International Solar Alliance (ISA)** framework to bid for utility-scale solar projects in the Uzbek desert regions.
- **Real-time Intelligence Sharing:** The Joint Working Group on Counter-Terrorism must establish secure communication channels for real-time intelligence sharing on terror financing, cyber threats, and cross-border narco-trafficking networks originating from the Af-Pak region.

Conclusion

Uzbekistan is a key partner in India's "Connect Central Asia" policy. India should focus on **timely implementation of projects, Digital Public Infrastructure (DPI), critical mineral cooperation and connectivity through Chabahar** to make the Comprehensive Strategic Partnership more effective. This can help India emerge as a **reliable, transparent and stabilising development partner in Central Asia** .

Drishti Mains Question:

"The success of India's Central Asia policy depends as much on economic connectivity as on diplomatic goodwill." Discuss.

Frequently Asked Questions (FAQs)

1. When did India and Uzbekistan establish diplomatic relations?

Diplomatic relations were established in **March 1992** , and bilateral ties were upgraded to a **Strategic Partnership in 2011** .

2. What is the significance of Uzbekistan for India's Central Asia policy?

Uzbekistan borders all other Central Asian republics and Afghanistan, making it a key **geopolitical pivot for India's Connect Central Asia policy** .

3. What are the major outcomes of the India-Uzbekistan partnership upgrade?

Cooperation has expanded to **defence, trade, critical minerals, uranium, clean energy, DPI, AI, healthcare and cultural heritage** .

4. What is the significance of UPI-UZQR integration?

It enables Indian **UPI applications to scan Uzbekistan's UZQR national codes** , strengthening digital connectivity and facilitating cross-border payments.

5. What are the major challenges in India-Uzbekistan relations?

Key challenges include **poor connectivity, banking barriers, China's economic dominance, Afghanistan-related security risks and non-tariff barriers** .

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UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. Consider the following countries: (2022)

1. Azerbaijan
2. Kyrgyzstan
3. Tajikistan
4. Turkmenistan
5. Uzbekistan

Which of the above have borders with Afghanistan?

- (a) 1, 2 and 5 only
- (b) 1, 2, 3 and 4 only
- (c) 3, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

Ans: (c)

Mains

Q. Critically analyse India's evolving diplomatic, economic, & strategic relations with the Central Asian Republics (CARs), highlighting their increasing significance in regional & global geopolitics. **(2024)**

Q . A number of outside powers have entrenched themselves in Central Asia, which is a zone of interest to India. Discuss the implications, in this context, of India's joining the Ashgabat Agreement, 2018. **(2018)**

Regulation of Social Media Addiction Among Children



For Prelims: Artificial Intelligence , Digital Personal Data Protection Act (DPDP Act), 2023 , Article 19 , Section 79 of the IT Act, 2000

For Mains: Social Media Regulation and Governance, Protecting children from online exploitation, addiction and targeted advertising, Intermediary Liability

Why in News?

Meta Platforms has reached a landmark settlement of up to USD 17.1 billion with multiple US states , agreeing to **new safety restrictions for users under 18** on Facebook and Instagram, following allegations that its platforms encouraged **addictive and compulsive use among children and teenagers** and violated child-privacy norms.

- The settlement is significant for India as the **Economic Survey 2025-26** has flagged **digital addiction as a growing public health concern** , particularly among the **15-24 age group** , while the Government is examining **age restrictions, usage limits and stronger parental-consent mechanisms** for minors.
- **The US settlement offers India a potential regulatory model** by combining **age assurance, daily usage caps, night-time restrictions, parental controls and independent oversight** , shifting greater responsibility from children and parents to **social-media platforms** themselves.

Summary

- **Social media addiction among children** is emerging as a major concern due to addictive platform design, algorithmic manipulation and excessive screen time, causing risks to **mental health, privacy, sleep, education and social development** .
- India needs a **rights-based, risk-sensitive regulatory framework** combining **child data protection, age verification, algorithmic accountability, safety-by-design, deepfake regulation and digital literacy** , while balancing **privacy, free speech and access to digital benefits** .

Why is Meta's \$17.1-Billion Settlement Significant?

Significance

- **Transition to Legally Binding Product Redesigns** : Moves child safety from voluntary, easily ignored features (e.g., "Take a Break") to court-enforceable product mandates including default 2-hour daily limits, midnight-to-6 AM lockouts, disabled cosmetic-procedure filters, and independent audit oversight.
- **Stronger Platform Accountability**: Shifts responsibility from parents and children towards **tech platforms** for preventing harmful and compulsive use.
- **Child Privacy Protection**: Highlights the importance of laws such as **US Children's Online Privacy Protection Act (COPPA) (1998)** in holding platforms accountable for collecting children's data without parental consent.
- **Industry-Wide Impact**: Could push other major platforms such as **YouTube and TikTok** to adopt similar safeguards.

Key Takeaways for India

- **Policy Blueprint:** Offers India a practical model for **age-based restrictions, daily usage limits, night-time controls and parental consent** .
- **Beyond Voluntary Regulation:** Supports India's debate on making **platforms legally responsible** for protecting minors rather than relying solely on parental supervision.
- **Need for Platform-Wide Rules:** Highlights the need for consistent safeguards across social-media platforms to prevent children from simply shifting to less-regulated apps.



How do Social Media Platforms "Hook" Children?

- **Dopamine Loops and Variable Rewards:** Social media platforms operate on the principles of the **Attention Economy** , utilizing behavioral psychology and neuroscience to maximize user engagement.
 - Similar to a slot machine, platforms use unpredictable "variable rewards" (likes, tags, comments, and notifications) to trigger the release of **dopamine** in the brain's mesolimbic system.
 - This creates a neurological dependency where the anticipation of a reward compels the user to continuously check their device.
- **Frictionless Design** : Features like infinite scrolling and auto-playing videos remove the brain's natural "stopping cues".
 - This bypasses conscious decision-making, leading to continuous, uninterrupted consumption.
- **Algorithmic Curation:** **Artificial Intelligence (AI)** tracks micro-interactions (like pause durations on a video) to curate hyper-personalized feeds. These adaptive algorithms prioritize engagement over safety, frequently pushing extreme, emotionally triggering, or highly idealized content.
- **Fear of Missing Out (FOMO):** The design exploits a child's primal need for social validation and peer inclusion. Features like "disappearing stories" or read receipts create artificial urgency, making adolescents feel they must remain constantly connected.

Impacts of Social Media Addiction on Children

- **Mental Health:** Increases **anxiety, depression, loneliness, low self-esteem, and body-image issues** , especially among teenage girls.
 - 2024 meta-analysis covering 143 studies and over one million adolescents found that higher platform **engagement directly correlates with heightened rates of depression and anxiety.**

- **Cognitive & Brain Effects:** Excessive use may affect the **prefrontal cortex** (it manages high-level thinking, decision-making, and emotional control) **and amygdala** (an area of your brain responsible for emotional processing), contributing to reduced attention, impulse control, and reward sensitivity.
- **Sleep & Physical Health:** Late-night scrolling and blue-light exposure disrupt **melatonin, sleep quality, and cognitive development** .
- **Withdrawal Effects:** Sudden restriction can cause **irritability, distress, and compulsive behaviour** in heavily dependent users.
- **Academic & Social Impact:** Excessive smartphone use can reduce **academic performance, real-world social interaction, and development of social skills** .

Global Responses:

- **International Bans:** Countries like Australia, Malaysia, and Indonesia have instituted strict bans or mandated parental controls for social media access for children under 16.
- **South Korea:** The **Cinderella/Shutdown Law** restricted online gaming for under-16s from midnight to 6 AM. It was abolished in 2021.
- **US: Children's Online Privacy Protection Act (COPPA) (1998)** requires parental consent for collecting data from under-13s.
 - **Children's Internet Protection Act (CIPA) (2000)** requires federally funded schools and libraries to filter harmful online content.
- **EU: General Data Protection Regulation (GDPR), 2018** provides strong data-privacy protections. The EU has also pursued stronger age and parental-consent safeguards for minors.
- **UK: Age-Appropriate Design Code** requires platforms to prioritise children's privacy and safety through stronger default settings.
- **France:** A 2023 law requires **parental authorisation for under-15s** to access social media, with penalties for non-compliance.
- **China:** Limits minors' internet use by age, with **daily time limits and nighttime restrictions** .

Indian Framework:

- **Digital Personal Data Protection Act (DPDP Act), 2023** : Regulates the processing of **children's personal data (under 18)** by requiring:
 - **Parental Consent:** Verifiable consent from a parent/guardian.
 - **Child Well-being:** Data processing must protect the child's welfare.
 - **No Tracking & Targeted Ads:** Restricts behavioural monitoring, tracking and targeted advertising.
- **Information Technology (IT) Rules, 2021:** Places the onus on social media intermediaries to ensure a safe online environment, though it relies heavily on self-regulation.

What are the Challenges in Regulating Social Media?

- **Age Verification Problem:** Self-declared age limits are easily bypassed, while **KYC or biometric verification** can create risks of surveillance, privacy violations and data breaches.
- **Platform Profit vs Child Safety:** Social-media platforms depend on **user engagement and screen time** , creating a conflict between their revenue model and efforts to curb addiction.
- **Digital Literacy Gap:** Many parents lack the skills to use **parental controls and privacy settings** , making age restrictions difficult to implement effectively.
- **Freedom of Speech vs Censorship:** Regulation must balance [Article 19\(1\)\(a\)](#) with [restrictions under Article 19\(2\)](#).
 - In **Anuradha Bhasin vs. Union of India (2020)**, the **Supreme Court** ruled that the Freedom of Speech and Expression and the Freedom to Practice any Profession over the medium of the Internet is a constitutionally protected Fundamental Right under **Article 19**.
 - Vague definitions of harmful content can cause **over-censorship and a chilling effect** on legitimate expression.
- **Privacy vs Traceability:** The **IT Rules, 2021** require certain messaging platforms to identify the **first originator** of information. This raises concerns that traceability could weaken **end-to-end encryption and privacy** .
- **Safe Harbour Dilemma:** **Section 79 of the IT Act, 2000** provides intermediary protection, but platforms increasingly **recommend, curate and monetise content** , blurring the line between passive intermediaries and active publishers.
- **Algorithmic Opacity:** Engagement-based algorithms can amplify **misinformation, hate speech and sensational content** , while proprietary algorithms remain difficult for regulators to scrutinise.
- **Deepfake Challenge:** AI-generated **deepfakes** , **voice clones and synthetic identities** make rapid detection and authentication of online content increasingly difficult.
- **Cross-Border Enforcement:** Foreign-based platforms and overseas data storage create difficulties in **evidence collection, content removal and law enforcement** , particularly when jurisdictions have conflicting laws.
- **Scale & Linguistic Diversity:** Billions of posts and India's **regional languages and dialects** make automated moderation prone to errors, especially in distinguishing **sarcasm, context and legitimate criticism** .
- **Reactive Regulation:** Reliance on **notice-and-takedown mechanisms** often means action is taken only after harmful content has spread. India's fragmented regulatory framework also creates **coordination gaps** .
- **Blanket Ban vs Digital Benefits:** Strict restrictions may protect children from online harms but can also limit **learning, creativity, socialisation, access to information and digital-skills development** .

How Can Social Media Regulation Be Strengthened?

- **Adopt Graded Age-Based Safeguards:** Introduce different levels of protection for 8-12, 12-16 and 16-18 age groups, recognising differences in maturity and vulnerability.
 - Consider **age-appropriate screen-time limits and night-time restrictions** , while allowing reasonable parental flexibility.
- **Modernise Legal Framework:** Replace the **IT Act, 2000** with the proposed **Digital India Act (DIA)** and shift from reactive **notice-and-takedown** to preventive *ex-ante* regulation.

- **Tiered Safe Harbour:** Hold platforms that **actively curate, amplify and monetise content** to higher accountability standards than passive intermediaries.
- **Independent Digital Regulator:** Establish an **autonomous, quasi-judicial authority** to oversee platforms, handle appeals and protect constitutional rights.
- **Safety by Design:** Mandate **algorithmic audits** , restrict addictive **dark patterns** like infinite scroll and autoplay, and make child-safety settings the default.
- **Deepfake Regulation:** Require **AI-generated content watermarking and provenance metadata** to improve identification of synthetic media.
- **Privacy-Preserving Enforcement:** Use tools such as **Zero-Knowledge Proofs and metadata analysis** instead of weakening end-to-end encryption, with surveillance subject to **judicial oversight and proportionality** .
- **International Cooperation:** Align with frameworks such as the **EU Digital Services Act (DSA)** and strengthen **MLATs** for faster cross-border access to digital evidence.
- **Digital & Media Literacy:** Integrate **digital wellness, misinformation detection, source verification, algorithmic awareness and responsible social-media use** into school education.
 - **Punjab’s introduction of AI as a mainstream subject from Classes I-XII** can serve as a model for expanding such digital literacy to include **mandatory training on safe and responsible social-media use** .
- **Institutional Support:** Expand specialised centres like **AIIMS Delhi’s Behavioural Addictions Clinic and CAR-AB** for early detection, counselling and treatment of digital addiction.

Conclusion

India needs to move from a **binary approach of unrestricted access vs blanket bans** towards a **rights-based, risk-sensitive regulatory model** that combines **child safety, privacy, free speech, digital literacy and platform accountability** . The focus should be on making digital platforms **safer by design rather than simply restricting access** .

Drishti Mains Question:

The regulation of social media for children involves a three-way trade-off between safety, privacy and freedom of expression. Examine.

Frequently Asked Questions (FAQs)

1. What does the DPDP Act, 2023 provide for children's data protection?

It requires **verifiable parental consent** for processing children's data and restricts **tracking, behavioural monitoring and targeted advertising** directed at children.

2. What is the key challenge with age verification on social media platforms?

Self-declared age is easily bypassed, while **biometric/KYC-based verification** can create risks of surveillance, privacy violations and data breaches.

3. What is safe harbour under Section 79 of the IT Act, 2000?

It provides **conditional intermediary liability protection** for third-party content, but platform curation, amplification and monetisation increasingly challenge the traditional passive-intermediary model.

4. How do social media algorithms contribute to digital addiction?

Personalised recommendations, variable rewards, infinite scrolling and autoplay maximise engagement by exploiting behavioural patterns and reducing natural stopping cues.

5. What regulatory approach should India adopt for social media platforms?

India should pursue **risk-sensitive, rights-based regulation** combining algorithmic audits, safety-by-design, privacy protection, child safeguards, platform accountability and digital literacy rather than relying on blanket bans.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/_S2JSGitrD4](https://www.youtube.com/embed/_S2JSGitrD4)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims:

Q1. 'Right to Privacy' is protected under which Article of the Constitution of India? (2021)

- (a) Article 15
- (b) Article 19
- (c) Article 21
- (d) Article 29

Ans: (c)

Q2. Right to Privacy is protected as an intrinsic part of Right to Life and Personal Liberty. Which of the following in the Constitution of India correctly and appropriately imply the above statement? (2018)

- (a) Article 14 and the provisions under the 42nd Amendment to the Constitution.
- (b) Article 17 and the Directive Principles of State Policy in Part IV.
- (c) Article 21 and the freedoms guaranteed in Part III.
- (d) Article 24 and the provisions under the 44th Amendment to the Constitution.

Ans: (c)

Mains:

Q. Examine the scope of Fundamental Rights in the light of the latest judgement of the Supreme Court on Right to Privacy. (2017)

National Animal Disease Control Programme (NADCP)



Source: PIB

Why in News?

The Government has highlighted the achievements of the **National Animal Disease Control Programme (NADCP)** in strengthening India's livestock health ecosystem through mass vaccination, disease surveillance and digital livestock management.

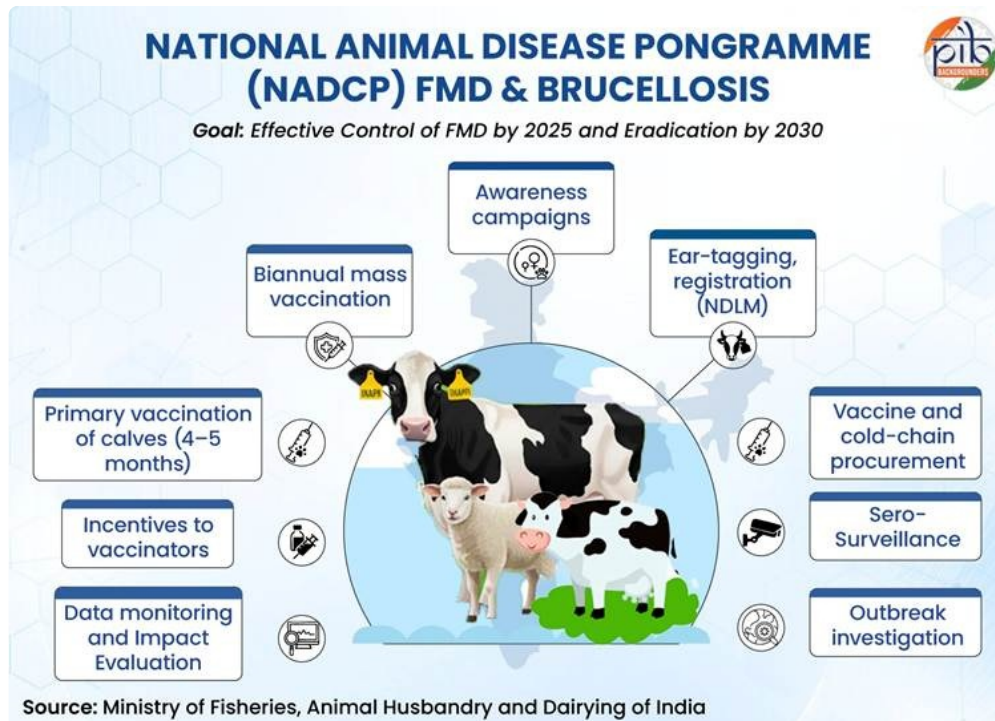
- The programme has administered **147.16 crore FMD vaccine doses** , benefiting **8.04 crore farmers** . As a result, reported **Foot and Mouth Disease (FMD)** outbreaks declined from **132 in 2019 to 40 in 2025** , while **Brucellosis** outbreaks reduced from **22 in 2019 to 15 in 2025** .

What are the Key Facts About National Animal Disease Control Programme (NADCP)?

- **About:** The **National Animal Disease Control Programme (NADCP)** is a **flagship Scheme launched in September 2019** to control and eradicate **Foot and Mouth Disease (FMD)** and **Brucellosis** through nationwide vaccination.
- **Vaccination Targets under NADCP:**
 - **100% vaccination of cattle, buffalo, sheep, goat, and pig populations against FMD.**
 - **100% vaccination of female bovine calves aged 4-8 months against Brucellosis.**
- **Disease Eradication Goal:** FMD outbreaks have declined by **more than 60% compared to the period before NADCP** . India continues efforts through vaccination, surveillance, animal identification and disease monitoring to achieve **FMD eradication by 2030** .
- **Funding Pattern:** It is a **Central Sector Scheme** under which **100% funding is provided by the Central Government to States/UTs** .
- **Disease Surveillance & Monitoring:** The programme includes **sero-surveillance, sero-monitoring, outbreak investigation, virus identification and impact assessment** to track disease prevalence and vaccination effectiveness.
- **Animal Identification & Data Management:** Livestock are **ear-tagged, registered and digitally recorded on the Bharat Pashudhan Portal** for effective monitoring and disease management.
- **Strengthening Delivery Systems:** NADCP supports **vaccine procurement, cold-chain infrastructure, awareness campaigns, official training, animal movement regulation and**

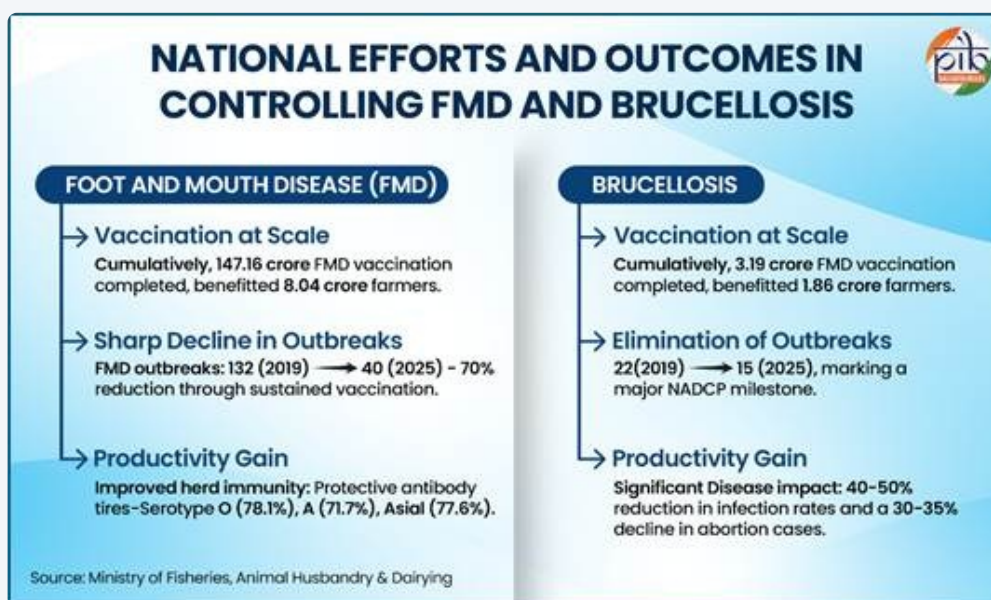
quarantine measures .

- **Financial Support:** The programme provides remuneration to vaccinators, with a minimum support of **₹3 per vaccination dose** , to ensure effective field-level implementation.
- **Integrated Livestock Health Framework:** Under the revised **Livestock Health and Disease Control Programme (LHDCP)** , NADCP, Livestock Health & Disease Control (LH&DC) component and **Pashu Aushadhi initiative** have been brought under one umbrella framework. The **2026-27 budget allocation for LHDCP is ₹2,010 crore** .



Major Livestock Diseases

- **Foot and Mouth Disease (FMD):** A highly contagious viral disease affecting cloven-hoofed animals such as cattle, buffaloes, sheep, goats and pigs. It causes **reduced milk yield, slower growth, reproductive problems and trade restrictions**.
- **Brucellosis:** A bacterial reproductive disease caused by *Brucella abortus* that affects fertility, lactation and overall productivity of cattle and buffaloes. Vaccination is the key preventive measure.



How is India Strengthening Institutional and Infrastructure Support for Animal Disease Control?

- **Research and Surveillance Support:** Financial assistance is provided to key **ICAR institutions** for controlling Foot and Mouth Disease (FMD) and Brucellosis. It supports vaccine research, quality testing, disease surveillance, sero-surveillance, disease modelling and State-wise sampling.
 - These measures facilitate the programme's scientific and evidence-based implementation.
- **Centralised Vaccine Procurement and Private-Sector Engagement:** States and Union Territories receive assistance to procure vaccination accessories and strengthen cold-chain infrastructure. The **Animal Husbandry Infrastructure Development Fund (AHIDF)** supports eligible entrepreneurs and private companies in establishing veterinary vaccine and drug-manufacturing facilities.
- **Veterinary Infrastructure and Service Delivery:** Veterinary facilities and **Mobile Veterinary Units (MVUs)** are being expanded to provide doorstep animal healthcare services through the toll-free number **1962**. Currently, **4,019 MVUs are operational across 29 States and Union Territories**, benefiting **136 lakh farmers** and providing treatment to **276 lakh animals**.
- **Bharat Pashudhan Portal:** The livestock portal assigns each animal a unique **12-digit Tag ID** and integrates registration with **FMD and Brucellosis vaccination records**. By **July 2026**, over **39 crore livestock** were registered, enabling real-time disease surveillance and data-driven livestock

management.

- **Awareness and Farmer Engagement: Multi-Purpose Artificial Insemination Technicians in Rural India (MAITRIs)** facilitate farmer outreach and doorstep livestock services. So far, **39,810 MAITRIs** have been trained with financial assistance.

National One Health Mission

- **One Health Approach** : Recognises the interconnectedness of **human, animal, and environmental health** promoting coordinated action across these sectors to address emerging health challenges.
- **National One Health Mission** : Approved by the **Prime Minister's Science, Technology and Innovation Advisory Council (PM-STIAC) in July 2022**, it is a cross-ministerial initiative aimed at coordinating, integrating, and strengthening existing One Health activities in India.
- **NDDB's Brucellosis Control Model**: Highlights the importance of **Vaccination of animals** , **Safe disposal of animal waste** , **Strong biosecurity measures** , **public awareness** and **Inter-departmental coordination**.



Pandemic Fund Project

- **About:** In **2024** , the Government launched the **Pandemic Fund Project on "Animal Health Security Strengthening in India for Pandemic Preparedness and Response."** It is a **US\$25 million** initiative supported by the **G20 Pandemic Fund** .
 - The project is implemented in partnership with the **Asian Development Bank (ADB)** , the **Food and Agriculture Organization (FAO)** , and the **World Bank** .

- **Objective:** The project complements the **NADCP** by strengthening animal health systems to improve pandemic preparedness. It enhances disease surveillance, including **genomic and environmental surveillance** .
 - It also upgrades laboratory infrastructure and promotes cross-border collaboration. These measures improve early detection and timely response to zoonotic and transboundary animal diseases.
- **Key Documents Released under the Project:**
 - **Standard Veterinary Treatment Guidelines (SVTG):** A comprehensive framework that promotes scientific veterinary practices, rational use of medicines, and antimicrobial stewardship. It also supports the [National Action Plan on Antimicrobial Resistance](#) .
 - **Crisis Management Plan (CMP) for Animal Diseases:** A national framework to strengthen preparedness and guide rapid response during disease outbreaks. It supports timely containment and effective recovery measures.

Conclusion

NADCP has transformed India's livestock health landscape through **mass vaccination, digital surveillance, and strengthened veterinary infrastructure** . By reducing FMD and Brucellosis burden, it has improved **livestock productivity, farmer incomes, food security, and pandemic preparedness** . Future integration of **One Health, genomic surveillance, AI-based disease monitoring, and global standards** will further strengthen India's vision of a **resilient and globally competitive livestock sector under Viksit Bharat 2047** .

Drishti Mains Question

Livestock health is critical for ensuring farmer income, food security and pandemic preparedness. Discuss.

Frequently Asked Questions (FAQs)

1. What is the National Animal Disease Control Programme (NADCP)?

NADCP is a government programme launched in 2019 to control and eradicate Foot and Mouth Disease (FMD) and Brucellosis through vaccination, surveillance and digital livestock management.

2. Which diseases are targeted under NADCP?

NADCP primarily targets: Foot and Mouth Disease (FMD) and Brucellosis.

3. What is Bharat Pashudhan Portal?

Bharat Pashudhan Portal is a digital livestock database that maintains animal identification, vaccination and health records.

4. What is the One Health approach?

One Health recognises the interconnection between human health, animal health and environmental health for managing disease risks.

5. How many livestock have been registered on Bharat Pashudhan Portal?

More than **39 crore livestock** have been registered on the portal.

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. As per the NSSO 70th Round “Situation Assessment Survey of Agricultural Households”, consider the following statements: (2018)

1. Rajasthan has the highest percentage share of agricultural households among its rural households.
2. Out of the total agricultural households in the country, a little over 60 percent belong to OBCs.
3. In Kerala, a little over 60 percent of agricultural households reported to have received maximum income from sources other than agricultural activities.

Which of the statements given above is/are correct?

- (a) 2 and 3 only
- (b) 2 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (c)

Mains

Q. Livestock rearing has a big potential for providing non-farm employment and income in rural areas. Discuss suggesting suitable measures to promote this sector in India. **(2015)**

India Signs Deal for 60 US Javelin Anti-Tank Missiles



Source: TH

Why in News?

India has signed a **Letter of Offer and Acceptance (LOA)** with the US for a **₹292-crore government-to-government deal** to acquire around **60 Javelin anti-tank guided missiles (ATGMs)** for the Indian Army through the **US Foreign Military Sales (FMS)** route.

- The acquisition will provide an **immediate operational capability** , while India's indigenous **Man-**

What are the Key Facts About the Javelin Missile System?

- **System Type:** It is a **man-portable, shoulder-fired, medium-range** anti-tank guided missile (ATGM) designed for individual or dismounted infantry use.
- **Fire-and-Forget Technology:** Utilizes a passive **imaging-infrared seeker** that automatically tracks targets post-launch, allowing troops to immediately relocate and evade counter-fire.
- **Soft Launch Capability:** Ejects the missile using an initial motor before main ignition, minimizing exhaust and enabling safe firing from **confined spaces** like bunkers and buildings.
- **Range & Weight:** Offers an effective operational range of **2.5 to 4 km** and weighs approximately **22 kg** , ensuring high mobility for infantry units.
- **Combat Proven:** Extensively **battle-tested** in high-intensity global conflicts, including deployments in Iraq, Afghanistan, and Ukraine.
- **Manufacturing Consortium:** Jointly produced by the **Javelin Joint Venture** (comprising US defence majors **Lockheed Martin** and **RTX**).
- **Strategic Synergy:** The procurement strengthens the **U.S.-India Major Defence Partnership** and paves the way for potential **mutually beneficial co-production** , which aligns with India's goal of bolstering its domestic defence industrial base.

Procurement Mechanism

- India's acquisition is executed under the **6 th tranche of Emergency Procurement (EP-6)**, which empowers the armed forces to fast-track purchases up to a ceiling of Rs 300 crore per equipment category.
- The deliveries are expected to begin by the end of the fiscal year 2026-27.
- The deal is aimed at plugging critical anti-armour and bunker-busting operational gaps in forward sectors following **Operation Sindoor**.



Man-Portable ATGM (MPATGM)

- The **indigenously developed MPATGM** is intended to replace the ageing **Milan-2T and Konkurs** systems in the Indian Army. India had also made an **emergency purchase of limited Spike-LR ATGMs in 2019** to address an immediate operational gap.
- In **January 2026** , DRDO successfully flight-tested the **third-generation, fire-and-forget MPATGM with top-attack capability** against a moving target at **KK Ranges, Ahilya Nagar, Maharashtra** .
- The **MPATGM incorporates indigenous technologies** including an **Imaging Infrared (IIR) homing seeker, all-electric control actuation system, fire control system, tandem warhead, propulsion system and high-performance sighting system** .
- It can be **launched from a tripod or military vehicle launcher** , while its **IIR seeker enables day-and-night operations** and its warhead is designed to defeat **modern main battle tanks** .
- **Bharat Dynamics Limited (BDL)** and **Bharat Electronics Limited (BEL)** are the **Development-Cum-Production Partners (DcPP)** for the weapon system, supporting its transition towards induction into the Indian Army.

Frequently Asked Questions (FAQs)

1. What is the Javelin missile system?

Javelin is a **man-portable, shoulder-fired, medium-range ATGM** featuring **fire-and-forget technology** and an imaging-infrared seeker.

2. What is the significance of the Javelin acquisition for India?

The procurement provides **immediate anti-armour capability** while strengthening the **India-US Major Defence Partnership** .

3. What is the MPATGM?

MPATGM is an **indigenously developed third-generation, fire-and-forget ATGM** with **top-attack capability** , developed by DRDO to replace ageing Milan-2T and Konkurs systems.

4. What are the key technologies incorporated in MPATGM?

It incorporates an **IIR homing seeker, fire-control system, all-electric control actuation system, tandem warhead, propulsion system and high-performance sighting system** .

5. How does MPATGM support defence indigenisation?

Its indigenous development, with **BDL and BEL as Development-Cum-Production Partners** , strengthens domestic defence R&D, manufacturing capability and **Aatmanirbhar Bharat** .

UPSC Civil Services Examination Previous Year Question (PYQ)

Q1. What is “Terminal High Altitude Area Defense (THAAD)”, sometimes seen in the news? (2018)

- (a) An Israeli radar system
- (b) India’s indigenous anti-missile programme
- (c) An American anti-missile system
- (d) A defence collaboration between Japan and South Korea.

Ans: (c)

Q2. With reference to Agni-IV Missile, which of the following statements is/are correct? (2014)

1. It is a surface-to-surface missile.
2. It is fuelled by liquid propellant only.
3. It can deliver one-tonne nuclear warheads about 7500 km away.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (a)

Mains

Q. What is the significance of Indo-US defence deals over Indo-Russian defence deals? Discuss with reference to stability in the Indo-Pacific region. **(2020)**

Pradhan Mantri Formalisation of Micro Food Processing Enterprises Scheme



Source: PIB

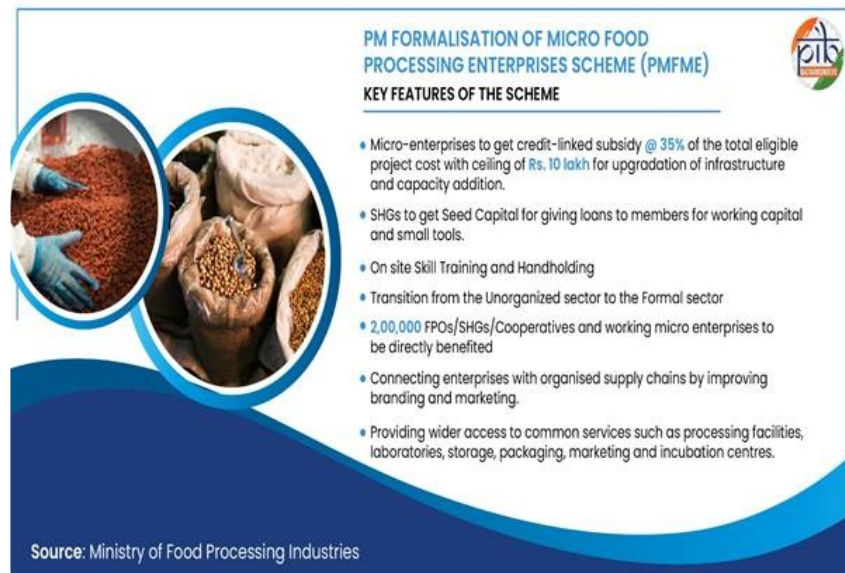
Why in News?

Recently, the **Ministry of Food Processing Industries** organised the **PMFME Bazaar** at India Habitat Centre, New Delhi, showcasing the achievements of micro food enterprises from across India.

- It highlighted the role of the **Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) Scheme** in supporting small food businesses through financial assistance, technical support, branding, and market linkages.

What are the Salient Features of PMFME Scheme?

- **About:** Launched on **29th June 2020**, the **Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) Scheme** is a **Centrally Sponsored Scheme** under the **Atmanirbhar Bharat Abhiyaan**.
- **Objective:** It aims to **formalise micro food processing enterprises, enhance their competitiveness, and support entrepreneurs** through financial, technical, and business assistance for setting up new units or upgrading existing ones.
- **Common Infrastructure Support:** The scheme provides **35% credit-linked capital subsidy** on eligible project costs up to a maximum of **₹3 crore** for **FPOs, FPCs, cooperatives, SHGs and their federations, and government agencies** establishing food processing facilities, value chain infrastructure, or incubation centres. Beneficiaries need to **contribute a minimum of 10% of the project cost**, with the remaining amount financed through bank loans.
- **Credit-Linked Subsidy for Individual Units:** Individuals, proprietorships, partnerships, FPOs, NGOs, cooperatives, SHGs, and private companies receive **35% credit-linked capital subsidy** for setting up new units or upgrading existing food processing units, subject to a maximum subsidy of **₹10 lakh per unit**.
- **One District One Product (ODOP) Approach:** PMFME adopts the **ODOP strategy** to promote economies of scale in procurement, common services, and marketing. The approach supports value chain development and infrastructure alignment, with ODOP approved for **713 districts across 35 States/UTs covering 137 unique products**.
- **Seed Capital Support for SHGs:** SHG members are provided **seed capital of ₹40,000 per member** for working capital and purchasing small tools, with a maximum limit of **₹4 lakh per SHG**. The grant is provided through SHG federations via **SRLM/SULM** for onward lending.
- **Marketing and Branding Support:** The scheme provides **50% financial assistance** for branding and marketing activities undertaken by groups of **FPOs, SHGs, cooperatives, or SPVs of micro food processing enterprises** to promote their processed food products.
- **Capacity Building and Training:** PMFME provides training through **Food Processing Entrepreneurship Development Programmes (FPEDP)** to beneficiaries, enabling them to improve technical skills, entrepreneurship capabilities, and business management.
- **Detailed Project Report (DPR) Support:** PMFME provides support for **preparation of DPRs related to branding and marketing proposals, covering product profile, quality control, packaging, pricing, promotion, storage, and marketing strategies**. Financial support up to **₹5 lakh** is available through State Nodal Agencies for DPR preparation.
- **Significance:** PMFME promotes **Atmanirbhar Bharat and Vocal for Local** by strengthening micro food enterprises, creating rural employment, improving value addition of local products, reducing wastage, and connecting traditional food products with wider markets.



Frequently Asked Questions (FAQs)

1. What was the objective of PMFME Bazaar?

The PMFME Bazaar aimed to provide a platform for micro food enterprises to showcase products, connect with buyers and explore new markets.

2. What is PMFME Scheme?

PMFME is a government scheme launched in 2020 to support micro food processing enterprises through financial assistance, capacity building, technology support and market linkage.

3. Which Ministry implements the PMFME Scheme?

The scheme is implemented by the **Ministry of Food Processing Industries**.

4. What subsidy is provided under PMFME?

Individual micro food processing units receive **35% credit-linked capital subsidy** Up to **₹10 lakh per unit**

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/LezciVjP1Ew](https://www.youtube.com/embed/LezciVjP1Ew)]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. With what purpose is the Government of India promoting the concept of “Mega Food Parks”? (2011)

1. To provide good infrastructure facilities for the food processing industry.
2. To increase the processing of perishable items and reduce wastage.
3. To provide emerging and eco-friendly food processing technologies to entrepreneurs.

Select the correct answer using the codes given below:

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Ans: (b)

Mains

Q . What are the reasons for the poor acceptance of a cost-effective small processing unit? How can the food processing unit be helpful to uplift the socioeconomic status of poor farmers? (2017)

Legal Metrology (Indian Standard Time) Rules, 2026



Source: PIB

The **Department of Consumer Affairs** has notified the **Legal Metrology (Indian Standard Time) Rules, 2026** , establishing **Indian Standard Time (IST)** as the mandatory, single time reference for all legal, administrative, commercial, and official purposes across India.

- **Statutory Implementation:** The rules will formally take effect 180 days from publication in the Official Gazette, providing government bodies, enterprises, and institutions a transitional window for system upgrades.
- **Statutory Backing:** The rules were notified under **Section 52 of the Legal Metrology Act, 2009** , giving legal and administrative significance to a uniform national time standard.

- **Core Objective:** Eliminates timing discrepancies across critical digital and physical infrastructure by standardizing accurate time-stamping for:
 - **Banking and digital payment transactions** to prevent reconciliation errors and fraud.
 - **Transport coordination** across railways, air traffic, and emergency response networks.
 - **Telecommunications, internet networks, and smart power grids** requiring precise microsecond-level synchronization.
 - **Legal records, administrative logs, and government filings** to ensure legal evidentiary validity.
- **Strategic Indigenisation:** Mandates the use of **NavIC (Navigation with Indian Constellation)** and domestic timing infrastructure to systematically reduce dependency on foreign satellite systems (such as the US GPS).
- **Institutional Dissemination Network:** Precise IST will be disseminated through domestic infrastructure backed by **CSIR-National Physical Laboratory (CSIR-NPL)** and **Legal Metrology laboratories**.
- **One Nation, One Time Initiative:** The new rules are part of the government's broader '**One Nation, One Time**' initiative, which seeks to establish a **unified national time standard** across critical infrastructure.
 - It uses **White Rabbit Technology (WRT)**, commissioned at the **Regional Reference Standard Laboratory (RRSL), Bengaluru**, to achieve **sub-nanosecond-level time synchronisation** across institutions such as **ISRO, SEBI, NSE and BSNL**.

Indian Standard Time (IST)

- **Longitudinal Basis:** IST is calculated on the basis of the **82.5° E longitude** (Standard Meridian).
 - The 82.5° E meridian passes through five Indian states: **Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh**.
- **Global Offset:** IST is exactly **5 hours and 30 minutes ahead** of Coordinated Universal Time (UTC +5:30).
- **The Official Timekeeper:** The **CSIR-National Physical Laboratory (NPL)** in New Delhi is the custodian of IST. It maintains the national time standard using a bank of highly precise Cesium and Rubidium atomic clocks, which are synchronised with the **International Bureau of Weights and Measures (BIPM) in France**.

Read more: [White Rabbit Technology- Based IST Dissemination Demonstration Network](#)

Lake Ontario



Recently, the US President signed an executive order to **rename Lake Ontario as “Lake America”** , citing its significance as a US asset and part of American heritage.

- Canada opposed the move, stating that the name **Ontario has Indigenous Wendat origins and is over 400 years old** , predating both the Canadian Confederation and the US Declaration of Independence.
- The **new name is not binding on Canada or other countries** as there is **no single global authority for standardising geographical names** .

Lake Ontario

- **Location and Identity:** Lake Ontario is one of the **five Great Lakes of North America** (Superior, Michigan, Huron, Erie, and Ontario) and lies between the **Ontario province of Canada and New York State of the USA** . It is the **only Great Lake that does not border Michigan** .
- **Size:** Lake Ontario is the **smallest of the five Great Lakes by surface area** and forms the **easternmost and final lake of the Great Lakes system** .
 - The **Great Lakes system contains around 20% of the world’s surface freshwater** , making it one of the largest freshwater ecosystems globally.
- **Connectivity and Navigation:** Its waters flow into the **Saint Lawrence River** , which connects the Great Lakes with the **Atlantic Ocean** , making it an important corridor for **navigation, trade, and freshwater exchange** .
 - It is connected to **Lake Erie through the Welland Canal** and forms part of the **Great Lakes-Saint Lawrence navigation system** .
- **Major Inflow:** The **Niagara River** is the principal source of water for Lake Ontario, contributing about **85% of its inflow** .
- **Ecological Importance:** The lake supports diverse aquatic ecosystems, including species such as the **Lake Sturgeon** , which faces conservation challenges.



Read More: [Arctic and Great Lakes Ice Trends](#)

